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United States Patent	12412016
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Wang; Chung-Ming et al.

Method and structure for mandrel and spacer patterning

Abstract

A method of fabricating a semiconductor device includes generating at least one photomask based on a layout and forming a plurality of active patterns on a substrate, using the at least one photomask. The layout includes a plurality of first patterns that extend parallel to each other in a first direction on a low-density region of the layout and a plurality of second patterns that extend parallel to each other in the first direction on a high-density region of the layout. The forming of the plurality of active patterns includes using the first patterns and the second patterns of the layout to respectively form a plurality of first active patterns and a plurality of second active patterns on the substrate.

Inventors: Wang; Chung-Ming (Chiayi, TW), Peng; Chih-Hsiung (Miaoli County, TW), Chang; Chi-Kang (New Taipei, TW), Chen; Kuei-Shun (Hsinchu, TW), Fu; Shih-Chi (Hsinchu County, TW)

Applicant: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsin-Chu, TW)

Family ID: 57629774

Assignee: TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.
(Hsinchu, TW)

Appl. No.: 17/720062

Filed: April 13, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220237356 A1	Jul. 28, 2022

Related U.S. Application Data

continuation parent-doc US 17229736 20210413 US 11748540 child-doc US 17720062
continuation parent-doc US 16725082 20191223 US 11010526 20210518 child-doc US 17229736

Publication Classification

Int. Cl.: **G06F30/39** (20200101); **H10D62/10** (20250101); **H10D84/01** (20250101); **H10D84/03** (20250101); **H10D84/83** (20250101); **H10D89/10** (20250101); G06F30/398 (20200101); G06F111/20 (20200101)

U.S. Cl.:

CPC **G06F30/39** (20200101); **H10D62/116** (20250101); **H10D84/0158** (20250101); **H10D84/038** (20250101); **H10D84/834** (20250101); **H10D89/10** (20250101); G06F30/398 (20200101); G06F2111/20 (20200101)

Field of Classification Search

USPC: 257/401

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Primary Examiner: Aisaka; Bryce M

Attorney, Agent or Firm: HAYNES AND BOONE, LLP

Background/Summary

PRIORITY (1) This is a continuation of U.S. patent application Ser. No. 17/229,736, filed Apr. 13, 2021, which is a continuation of U.S. patent application Ser. No. 16/725,082, filed Dec. 23, 2019, now issued U.S. Pat. No. 11,010,526, which is a continuation of U.S. patent application Ser. No. 15/945,356, filed Apr. 4, 2018, now issued U.S. Pat. No. 10,521,541, which is a divisional of U.S. patent application Ser. No. 14/801,383, filed Jul. 16, 2015, now issued U.S. Pat. No. 9,946,827, herein incorporated by reference in their entireties.

BACKGROUND

(1) As integrated circuit (IC) technologies are continually progressing to smaller technology nodes, such as a 32 nm technology node and below, simply scaling down similar designs used at larger

nodes often results in inaccurate or poorly shaped device features due to the resolution limit of conventional optical lithography technology. Examples of inaccurate or poorly shaped device features include rounding, pinching, necking, bridging, dishing, erosion, metal line thickness variations, and other characteristics that affect device performance. One approach to improving image printing quality on a wafer is to use restrictive design rules (RDR) in IC layout designs. An exemplary IC layout according to RDR includes parallel line patterns extending in the same direction and spaced by a pattern pitch. The line width and pattern pitch are designed so as to improve image printing quality by utilizing constructive light interference.

(2) However, in a large scale IC, not all patterns are designed according to the same design rules. For example, an IC may include both logic circuits and embedded static random-access memory (SRAM) cells. The SRAM cells may use smaller pitches for area reduction, while the logic circuits may use larger pitches. For another example, an IC may include multiple off-the-shelf macros, each of which has been laid out according to its own set of RDRs. In such ICs, multiple layout blocks may be used. Each layout block is designed according to a set of RDRs and different layout blocks may use different RDRs. A space is provided between any two layout blocks to accommodate printing inaccuracy such as line end rounding, as well as to meet certain spacing requirements for IC manufacturing. This space becomes a concern when greater device integration is desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when they are read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a simplified block diagram of an embodiment of an integrated circuit (IC) manufacturing system and an associated IC manufacturing flow.

(3) FIG. 2 is a more detailed block diagram of the mask house shown in FIG. 1 according to various aspects of the present disclosure.

(4) FIG. 3 shows a high-level flowchart of a method of manufacturing an IC according to various aspects of the present disclosure.

(5) FIGS. 4 and 5 illustrate an IC having two layout blocks in accordance with some embodiments.

(6) FIGS. 6A, 6B, 6C, 6D, and 6E illustrate an IC design layout modified according to the method shown in FIG. 3, in accordance with an embodiment.

(7) FIG. 7 illustrates a flowchart of a method of patterning a substrate according to various aspects of the present disclosure.

(8) FIGS. 8, 9A, 9B, 9C, 10A, 10B, 10C, 11A, 11B, 11C, 12A, 12B, 12C, 13A, 13B, 13C, 14A, 14B, and 14C illustrate top view and/or cross-sectional views of an IC in various manufacturing steps of the method in FIG. 7, in accordance with some embodiment.

DETAILED DESCRIPTION

(9) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of

simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(10) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(11) FIG. 1 is a simplified block diagram of an embodiment of an IC manufacturing system **100** and an IC manufacturing flow associated therewith, which may benefit from various aspects of the provided subject matter. The IC manufacturing system **100** includes a plurality of entities, such as a design house **120**, a mask house **140**, and an IC manufacturer **160** (i.e., a fab), that interact with one another in the design, development, and manufacturing cycles and/or services related to manufacturing an IC device **162**. The plurality of entities are connected by a communications network, which may be a single network or a variety of different networks, such as an intranet and the Internet, and may include wired and/or wireless communication channels. Each entity may interact with other entities and may provide services to and/or receive services from the other entities. One or more of the design house **120**, mask house **140**, and IC manufacturer **160** may be owned by a single larger company, and may even coexist in a common facility and use common resources.

(12) The design house (or design team) **120** generates an IC design layout **102**. The IC design layout **102** includes various geometrical patterns designed for the IC device **162**. An exemplary IC design layout **102** is shown in FIG. 4, which includes two layout blocks **104** and **106** separated by a space **108**. The two layout blocks **104** and **106** each include a plurality of patterns **110** and **112** respectively, designed according to some RDRs. Particularly, the patterns **110** and **112** are line patterns oriented lengthwise along the X direction. The line patterns **110** each have a line width W_1 and are spaced by an edge-to-edge pitch $P_{sub.1}$ along the Y direction that is orthogonal to the X direction. The line patterns **112** each have a line width $W_{sub.2}$ and are spaced by an edge-to-edge pitch $P_{sub.2}$ along the Y direction. The various geometrical patterns in the IC design layout **102**, such as the line patterns **110** and **112**, may correspond to patterns of metal, oxide, or semiconductor layers that make up various components of the IC device **162** to be fabricated. The various components may include active regions, gate electrodes, metal lines or vias of an interlayer interconnection, and openings for bonding pads, which are to be formed in a semiconductor substrate (such as a silicon wafer) and various material layers disposed on the semiconductor substrate. In an embodiment, the line patterns **110** and **112** are mandrel patterns used in a mandrel-spacer double patterning process for improving pattern density, which will be described in more details later. The design house **120** implements a proper design procedure to form the IC design layout **102**. The design procedure may include logic design, physical design, and/or place and route. The IC design layout **102** is presented in one or more data files having information of the geometrical patterns. For example, the IC design layout **102** can be expressed in a GDSII file format, a DFII file format, or another suitable computer-readable data format.

(13) The mask house **140** uses the IC design layout **102** to manufacture one or more masks to be used for fabricating various layers of the IC device **162**. The mask house **140** performs mask data preparation **132**, mask fabrication **144**, and other suitable tasks. The mask data preparation **132** translates the IC design layout **102** into a form that can be physically written by a mask writer. The mask fabrication **144** then fabricates a plurality of masks that are used for patterning a substrate (e.g., a wafer). In the present embodiment, the mask data preparation **132** and mask fabrication **144** are illustrated as separate elements. However, the mask data preparation **132** and mask fabrication **144** can be collectively referred to as mask data preparation.

(14) In the present embodiment, the mask data preparation **132** includes a dummy mandrel insertion operation, which inserts dummy line patterns in the space **108** (FIG. **4**) so as to improve pattern density and to reduce the area needed by the space **108**. This will be described in details later. Further in the present embodiment, the mask data preparation **132** prepares a mandrel pattern layout and a cut pattern layout to be used in a spacer double patterning process. The mandrel pattern layout defines a mandrel pattern in a first exposure and the cut pattern layout defines a cut pattern in a second exposure. The cut pattern removes unwanted portions of the mandrel pattern, a derivative, or both. The final pattern includes the mandrel pattern plus the derivative but not the cut pattern.

(15) The mask data preparation **132** may further include optical proximity correction (OPC) which uses lithography enhancement techniques to compensate for image errors, such as those that can arise from diffraction, interference, or other process effects. The mask data preparation **132** may further include a mask rule checker (MRC) that checks the IC design layout with a set of mask creation rules which may contain certain geometric and connectivity restrictions to ensure sufficient margins, to account for variability in semiconductor manufacturing processes, etc. The mask data preparation **132** may further include lithography process checking (LPC) that simulates processing that will be implemented by the IC manufacturer **160** to fabricate the IC device **162**. The processing parameters may include parameters associated with various processes of the IC manufacturing cycle, parameters associated with tools used for manufacturing the IC, and/or other aspects of the manufacturing process.

(16) It should be understood that the above description of the mask data preparation **132** has been simplified for the purposes of clarity, and data preparation may include additional features such as a logic operation (LOP) to modify the IC design layout according to manufacturing rules. Additionally, the processes applied to the IC design layout **102** during data preparation **132** may be executed in a variety of different orders.

(17) After mask data preparation **132** and during mask fabrication **144**, a mask or a group of masks are fabricated based on the modified IC design layout. For example, an electron-beam (e-beam) or a mechanism of multiple e-beams is used to form a pattern on a mask (photomask or reticle) based on the modified IC design layout. The mask can be formed in various technologies such as a transmissive mask or a reflective mask. In an embodiment, the mask is formed using binary technology, where a mask pattern includes opaque regions and transparent regions. A radiation beam, such as an ultraviolet (UV) beam, used to expose the image sensitive material layer (e.g., photoresist) coated on a wafer, is blocked by the opaque region and transmits through the transparent regions. In one example, a binary mask includes a transparent substrate (e.g., fused quartz) and an opaque material (e.g., chromium) coated in the opaque regions of the mask. In another example, the mask is formed using a phase shift technology. In the phase shift mask (PSM), various features in the pattern formed on the mask are configured to have proper phase difference to enhance the resolution and imaging quality. In various examples, the phase shift mask can be attenuated PSM or alternating PSM.

(18) The IC manufacturer **160**, such as a semiconductor foundry, uses the mask (or masks) fabricated by the mask house **140** to fabricate the IC device **162**. The IC manufacturer **160** is an IC fabrication business that can include a myriad of manufacturing facilities for the fabrication of a variety of different IC products. For example, there may be a manufacturing facility for the front end fabrication of a plurality of IC products (i.e., front-end-of-line (FEOL) fabrication), while a second manufacturing facility may provide the back end fabrication for the interconnection and packaging of the IC products (i.e., back-end-of-line (BEOL) fabrication), and a third manufacturing facility may provide other services for the foundry business. In the present embodiment, a semiconductor wafer **152** is fabricated using the mask (or masks) to form the IC device **162**. The semiconductor wafer **152** includes a silicon substrate or other proper substrate having material layers formed thereon. Other proper substrate materials include another suitable elementary

semiconductor, such as diamond or germanium; a suitable compound semiconductor, such as silicon carbide, indium arsenide, or indium phosphide; or a suitable alloy semiconductor, such as silicon germanium carbide, gallium arsenic phosphide, or gallium indium phosphide. The semiconductor wafer may further include various doped regions, dielectric features, and multilevel interconnects (formed at subsequent manufacturing steps). The mask may be used in a variety of processes. For example, the mask may be used in an ion implantation process to form various doped regions in the semiconductor wafer, in an etching process to form various etching regions in the semiconductor wafer, and/or other suitable processes.

(19) FIG. 2 is a more detailed block diagram of the mask house **140** shown in FIG. 1 according to various aspects of the present disclosure. In the illustrated embodiment, the mask house **140** includes a mask design system **180** that is tailored to perform the functionality described in association with mask data preparation **132** of FIG. 1. The mask design system **180** is an information handling system such as a computer, server, workstation, or other suitable device. The system **180** includes a processor **182** that is communicatively coupled to a system memory **184**, a mass storage device **186**, and a communication module **188**. The system memory **184** provides the processor **182** with non-transitory, computer-readable storage to facilitate execution of computer instructions by the processor. Examples of system memory may include random access memory (RAM) devices such as dynamic RAM (DRAM), synchronous DRAM (SDRAM), solid state memory devices, and/or a variety of other memory devices known in the art. Computer programs, instructions, and data are stored on the mass storage device **186**. Examples of mass storage devices may include hard drives, optical drives, magneto-optical drives, solid-state storage devices, and/or a variety other mass storage devices known in the art. The communication module **188** is operable to communicate information such as IC design layout files with the other components in the IC manufacturing system **100**, such as the design house **120**. Examples of communication modules may include Ethernet cards, 802.11 WiFi devices, cellular data radios, and/or other suitable devices.

(20) In operation, the mask design system **180** is configured to manipulate the IC design layout **102** before it is transferred to a mask **190** by the mask fabrication **134**. In an embodiment, the mask data preparation **132** is implemented as software instructions executing on the mask design system **180**. To further this embodiment, the mask design system **180** receives a first GDSII file **192** containing the IC design layout **102** from the design house **120**, and modifies the IC design layout **102**, for example, to insert dummy patterns and to perform other manufacturability enhancement. After the mask data preparation **132** is complete, the mask design system **180** transmits a second GDSII file **194** containing a modified IC design layout to the mask fabrication **134**. In alternative embodiments, the IC design layout may be transmitted between the components in IC manufacturing system **100** in alternate file formats such as DFII, CIF, OASIS, or any other suitable file type. Further, the mask design system **180** and the mask house **140** may include additional and/or different components in alternative embodiments.

(21) FIG. 3 is a high-level flowchart of a method **300** of manufacturing an IC according to various aspects of the present disclosure. In a brief overview, the method **300** includes operations **302**, **304**, **306**, **308**, and **310**. The operation **302** receives an IC design layout having multiple layout blocks separated by spaces. The operation **304** modifies the IC design layout by inserting dummy patterns to the spaces. The operation **306** outputs a mandrel pattern layout and a cut pattern layout for mask fabrication. The operation **308** fabricates a first mask with the mandrel pattern layout and a second mask with the cut pattern layout. The operation **310** patterns a substrate with the first mask and the second mask, for example, using a spacer patterning technique. The method **300** may be implemented in the various components of the IC manufacturing system **100**. For example, the operations **302**, **304**, and **306** may be implemented in the mask data preparation **132** of the mask house **140**; the operation **308** may be implemented in the mask fabrication **134** of the mask house **140**; and the operation **310** may be implemented in the IC manufacturer **160**. The method **300** is

merely an example for illustrating various aspects of the provided subject matter. Additional operations can be provided before, during, and after the method **300**, and some operations described can be replaced, eliminated, or moved around for additional embodiments of the method. The method **300** in FIG. **3** is a high-level overview and details associated with each operation therein will be described in association with the subsequent figures in the present disclosure.

(22) At operation **302**, the method **300** (FIG. **3**) receives the IC design layout **102** as shown in FIG. **4**. Referring to FIG. **4**, the IC design layout **102** includes various geometrical patterns for creating features of an IC. In the present embodiment, the IC design layout **102** includes the two layout blocks **104** and **106**. Each of the layout blocks **104** and **106** is a rectangular region and includes patterns conforming to some restricted design rules. Particularly, the layout block **104** includes the line patterns **110** having the line width $W_{sub.1}$ and the pattern pitch $P_{sub.1}$, and the layout block **106** includes the line patterns **112** having the line width $W_{sub.2}$ and the pattern pitch $P_{sub.2}$. The pattern pitches $P_{sub.1}$ and $P_{sub.2}$ are defined using edge-to-edge distance in the present embodiment. They may also be defined using center-line-to-center-line distance in alternative embodiments. The line patterns **110** and **112** are oriented lengthwise along the same direction (along the X direction), but the line widths $W_{sub.1}$ and $W_{sub.2}$ may be the same or different, and the pattern pitches $P_{sub.1}$ and $P_{sub.2}$ may be the same or different. In the present example, $P_{sub.1}$ is greater than $P_{sub.2}$ and $W_{sub.1}$ is greater than $W_{sub.2}$. The layout blocks **104** and **106** are separate for various reasons. For example, they may include different design macros or different types of circuit elements (e.g., logic circuits and SRAM cells). For another example, they may be designed to have different line widths and pitches so as to avoid accidental linking between the two blocks. Further, the layout blocks **104** and **106** are shown in rectangular regions for the purpose of simplification and they may be in other shapes or other polygons in various embodiments.

(23) In embodiments, the line patterns **110** and **112** may be used for creating IC features such as active regions, source and drain features, gate electrodes, metal lines or vias, and openings for bonding pads. In the present embodiment, the line patterns **110** and **112** define mandrel patterns, upon whose sidewalls a spacer will be formed and the spacer will be used for etching a substrate to form fins for fin field effect transistors (FinFETs). This will be described in greater details later.

(24) The layout blocks **104** and **106** are separated by the space **108**. In the present embodiment, the space **108** is also of a rectangular shape for simplification purposes. Further, in the present embodiment, the space **108** corresponds to a cut pattern **116**, as shown in FIG. **5**. The cut pattern **116** is used for removing features from a substrate. In a typical design, the space **108** is needed for meeting various manufacturing rules. For example, a manufacturing rule may set a minimum distance between one line end to another line end, such as between the ends of the line patterns **110** and the adjacent ends of the line patterns **112**. If the IC design layout **102** violates the manufacturing rule, a design rule checker (DRC) will flag a warning or an error so that the IC design layout may be modified or corrected before proceeding to the next fabrication stage (e.g., the mask fabrication **134** of FIG. **1**). For another example, when mandrel lines are formed on a substrate according to the line patterns **110** and **112** and spacers are formed on the sidewalls of the mandrel lines, a manufacturing rule may require that the spacers in the two separate layout blocks **104** and **106** do not come into contact with each other. For yet another example, due to the limitations of the conventional optical lithography technology, the ends of the line patterns **110** and **112** may become rounded after being printed on a wafer and the rounded ends may extend into the space **108**. A manufacturing rule may therefore require enough spacing between the line ends to account for the lithography inaccuracy.

(25) Due to the various concerns and other factors, the space **108** and the corresponding cut pattern **116** may be necessary for meeting manufacturability requirements in some instances. However, they typically take up large areas on a wafer. In one example, the width of the cut pattern **116** (along the X direction) is about 200 nanometers (nm) to about 300 nm in a 16 nm process node. This constitutes an added cost for the final IC devices **162**. Therefore, it is desirable to reduce the

space **108** thereby improving design density and reducing manufacturing costs. The provided subject matter addresses this issue, among others.

(26) At operation **304**, the method **300** (FIG. **3**) adds dummy mandrel patterns to the space **108** thereby connecting some of the line patterns **110** and some of the line patterns **112**. Referring to FIG. **6A**, shown therein is the IC design layout **102** with three dummy mandrel patterns **114A-C** inserted. Note that the number of the dummy mandrel patterns **114**, as well as their shape, width, and orientation, as shown in FIG. **6A**, are only for illustrative purposes and do not limit the provided subject matter. In embodiments, one line pattern **110** may be connected to one or more line patterns **112** by one or more dummy mandrel pattern **114**, and vice versa. In addition, not all line patterns **110** and **112** are connected by a dummy mandrel pattern. In the particular example shown in FIG. **6A**, line pattern **110A** is connected to two line patterns **112A** and **112B** through two dummy mandrel patterns **114A** and **114B** respectively; line pattern **110B** is connected to line pattern **112C** through dummy mandrel pattern **114C**; and line patterns **110C** and **112D** are not connected by any dummy mandrel patterns. Further in the present embodiment, the dummy mandrel patterns **114A-C** are linear pieces having about the same width as the line patterns **112**, and are each oriented lengthwise along a direction that may be the same as or different from the X direction. For example, the dummy mandrel pattern **114C** is oriented lengthwise in a direction U that forms an intersecting angle Θ with the direction X. In an embodiment, the angle Θ is limited to 45 degrees or less, such as 30 degrees or less, for manufacturability concerns. In another embodiment, the angle Θ may be limited to another range of values depending on the manufacturing process.

(27) The purposes and benefits of inserting the dummy mandrel patterns **114** are many folds and the following are not intended to be limiting. First, after connecting line patterns between two adjacent layout blocks, there is no longer a concern for violating rules about minimum gap between line ends for the connected line patterns within the space **108**. Second, there is no longer a concern for line end rounding issues for the connected line patterns within the space **108**. Third, when spacers are later formed in the layout blocks **104** and **106**, there is no long a concern for keeping the spacers separate in the space **108** because the connected line patterns (e.g., line patterns **110B** and **112C**) have become a continuous piece. These three aspects help reduce the size of the space **108**. In another word, the layout blocks **104** and **106** may be placed closer to each other than in conventional design flow without violating manufacturing rules. In one example, the width of the cut pattern **116** (FIG. **5**) may be reduced to about 100 nm or less in a 16 nm process node with the insertion of dummy mandrel patterns as provided in the present disclosure.

(28) There are additional benefits. For example, with the addition of the dummy mandrel patterns, spacer pattern density increases in the space **108**. This helps improve the shape and critical dimension of fins that are etched with the spacer. For example, the dummy mandrel patterns increase pattern density of the IC **102** and improve chemical mechanical planarization (CMP) loading effect during IC fabrication.

(29) In embodiments, a user may define criteria for where to insert the dummy mandrel patterns **114** and what shape, size, and orientation each dummy mandrel pattern **114** may take. FIGS. **6B-6E** illustrate some non-limiting examples. Referring to FIG. **6B**, a line pattern **110** is connected to three line patterns **112** by three dummy mandrel patterns **114A**, **114B**, and **114C**. The three dummy mandrel patterns are linear pieces that are oriented lengthwise along different directions.

Particularly, the dummy mandrel pattern **114B** is oriented lengthwise along the X direction, the dummy mandrel pattern **114A** is oriented lengthwise along a direction U1 that is different from the X direction, and the dummy mandrel pattern **114C** is oriented lengthwise along a direction U2 that is different from both the U1 and the X directions. Referring to FIG. **6C**, the line pattern **110** is connected to two line patterns **112A** and **112C** that are not adjacent. In some instances, line end rounding of a narrow mandrel (such as the line pattern **112B**) may be well controlled and therefore it is not necessary to connect every narrow mandrel using dummy mandrel patterns. Referring to

FIG. 6D, the dummy mandrel pattern **114** shown therein is not a linear piece. Instead, it has three linear sections, **114-1**, **114-2**, and **114-3**. For example, section **114-1** may be made by extending the line pattern **110** into the space **108**, section **114-3** may be made by extending the line pattern **112** into the space **108**, and section **114-2** connects the sections **114-1** and **114-3**. The lengths of the sections **114-1** and **114-3** may be adjusted so that the section **114-2** is oriented lengthwise in a particular direction. This may be advantageous in providing more uniform patterns in the IC design layout **102**. Various other embodiments of the dummy mandrel patterns are within the scope of the present disclosure. The IC design layout **102** in FIG. 6E is similar to that in FIG. 6D.

(30) At operation **306**, the method **300** (FIG. 3) outputs layout data for mask fabrication. In an embodiment, the layout data includes a mandrel pattern layout and a cut pattern layout. In the present embodiment, the mandrel pattern layout includes the line patterns received in operation **302** as well as the dummy mandrel patterns inserted in operation **304**; and the cut pattern layout includes one or more patterns corresponding to the space between layout blocks. One example of the mandrel pattern layout and the cut pattern layout is shown in FIG. 6E. Referring to FIG. 6E, the mandrel pattern layout for the IC design **102** includes the line pattern(s) **110** in the layout block **104**, the line pattern(s) **112** in the layout block **106**, and the dummy mandrel pattern(s) **114A** and **114B** inserted in the space **108**. The cut pattern layout for the IC design **102** includes a cut pattern **116** corresponding to the space **108**. In the present embodiment, the cut pattern layout of the IC design **102** further includes one or more cut patterns **118** which will remove spacers formed at the ends of the line patterns **110** and **112**. Further, each of the mandrel pattern layout and the cut pattern layout may also include certain assist features, such as those features for imaging effect, processing enhancement, and/or mask identification information. In embodiments, operation **306** outputs the mandrel pattern layout and the cut pattern layout in a computer-readable format for subsequent fabrication stage. For example, the layouts may be outputted in GDSII, DFII, CIF, OASIS, or any other suitable file format.

(31) At operation **308**, the method **300** (FIG. 3) manufactures a first mask with the mandrel pattern layout and manufactures a second mask with the cut pattern layout. Operation **308** may manufacture other masks for various layers and features of the IC **162**. In embodiments, the first mask and the second mask may be transmissive masks (e.g., for DUV lithography) or reflective masks (e.g., for EUV lithography), and may include imaging enhancement features such as phase shifting. In embodiments where maskless lithography, such as e-beam direct writing, is used, operation **308** is bypassed or involves data preparation for the particular direct writer without fabricating an actual mask.

(32) At operation **310**, the method **300** (FIG. 3) patterns a substrate (such as a wafer) with the first mask and the second mask to fabricate the final IC device **162**. The operation **310** involves a variety of lithography patterning and etching steps. An embodiment of the operation **310** (also referred to as the method **310**) is illustrated in FIG. 7, which uses a spacer technique in forming FinFETs. In various embodiments, operation **310** may pattern a substrate with or without using a spacer technique. The method **310** is merely an example for illustrating various aspects of the provided subject matter. Additional operations can be provided before, during, and after the method **310**, and some operations described can be replaced, eliminated, or moved around for additional embodiments of the method. The various operations in FIG. 7 are discussed below in conjunction with FIGS. 8-14C.

(33) At operation **352**, the method **310** (FIG. 7) deposits dielectric layers **804** and **806** over a substrate **802** (e.g., a semiconductor wafer) as shown in FIG. 8. The substrate **802** includes silicon in the present embodiment. In various embodiments, the substrate **802** may include another elementary semiconductor, such as germanium; a compound semiconductor, such as silicon carbide, indium arsenide, or indium phosphide; or an alloy semiconductor, such as silicon germanium carbide, gallium arsenic phosphide, or gallium indium phosphide. Materials suitable for the dielectric layers **804** and **806** include, but not limited to, silicon oxide, silicon nitride, poly-

silicon, Si.sub.3N.sub.4, SiON, TEOS, nitrogen-containing oxide, nitride oxide, high-k materials, or combinations thereof. The dielectric layers **804** and **806** are each formed by one or more deposition techniques, such as thermal oxidation, chemical vapor deposition (CVD), and physical vapor deposition (PVD).

(34) At operation **354**, the method **310** (FIG. 7) forms mandrel patterns in the dielectric layer **806**. Refer to FIGS. **9A**, **9B**, and **9C** collectively, which illustrate a portion of the IC device **162**. Particularly, FIG. **9A** shows a top view of the device **162** which includes a first region corresponding to the layout block **104** (FIG. **6E**) and a second region corresponding to the layout block **106** (FIG. **6E**). The device **162** further includes a third region sandwiched between the first and second regions. The third region corresponds to the space **108** of FIG. **6E**. For the convenience of discussion, the first region is also referred to as the region **104**, the second region is also referred to as the region **106**, and the third region is also referred to as the region **108**.

(35) The device **162** is patterned to have a mandrel pattern **806'** (a patterned dielectric layer **806**). The mandrel pattern **806'** includes multiple sections **806A-E** connected together: section **806A** corresponds to the line pattern **110A** in the layout block **104** (FIG. **6E**), sections **806B** and **806D** correspond to the dummy mandrel patterns **114A** and **114B** (FIG. **6E**) respectively, and sections **806C** and **806E** correspond to the line patterns **112A** and **112B** (FIG. **6E**) respectively. The sections **806A**, **806C**, and **806E** are oriented lengthwise along the X direction, while the sections **806B** and **806D** are each oriented lengthwise along a respective direction different from the X direction. FIG. **9B** is a cross-sectional view of the device **162** along the **1-1** line of FIG. **9A**. In the present example, the **1-1** line goes through the center lines of the sections **806B** and **806C**. Therefore it is not a straight line. FIG. **9C** shows cross-sectional views of the device **162** along the **2A-2A**, **2B-2B**, and **2C-2C** lines of FIG. **9A**. The mandrel pattern **806'** is a protruding feature (also known as a line pattern) in the present embodiment. In an alternative embodiment, the mandrel pattern **806'** may be a trench feature.

(36) The mandrel pattern **806'** is formed by patterning the dielectric layer **806** with a procedure including a lithography process and an etching process. For example, a photoresist (or resist) layer is formed on the dielectric layer **806** using a spin-coating process and soft baking process. Then, the photoresist layer is exposed to a radiation using the first mask manufactured in the operation **308** (FIG. **3**). The exposed photoresist layer is developed using post-exposure baking, developing, and hard baking thereby forming a patterned photoresist layer over the dielectric layer **806**. Subsequently, the dielectric layer **806** is etched through the openings of the patterned photoresist layer, forming the mandrel pattern **806'**. The etching process may include a dry (or plasma) etching, a wet etching, or other suitable etching methods. The patterned photoresist layer is removed thereafter using a suitable process, such as wet stripping or plasma ashing. During the above photolithography process, the density and regularity of the mandrel patterns (**110**, **112**, and **114**) help improve pattern critical dimension uniformity in view of optical proximity effect.

(37) At operation **356**, the method **310** (FIG. 7) forms a spacer **808**. Refer to FIG. **10A** (a top view of the device **162**), FIG. **10B** (a cross-sectional view of the device **162** along the **1-1** line of FIG. **10A**), and FIG. **10C** (cross-sectional views of the device **162** along the **2A-2A**, **2B-2B**, and **2C-2C** lines of FIG. **10A**). The spacer **808** is formed on sidewalls of the mandrel pattern **806'** and completely surrounds the mandrel pattern **806'**. In an embodiment, the spacer **808** has a substantially uniform thickness. The spacer **808** includes one or more material different from the mandrel pattern **806'**. In an embodiment, the spacer **808** may include a dielectric material, such as titanium nitride, silicon nitride, or titanium oxide. The spacer **808** can be formed by various processes, including a deposition process and an etching process. For example, the deposition process may include a CVD process or a PVD process. For example, the etching process may include an anisotropic etch such as plasma etch. In an embodiment of the method **310**, the operation **356** is bypassed and the mandrel pattern **806'** is used for etching the substrate without forming the spacer **808**.

(38) At operation **358**, the method **310** (FIG. 7) removes the mandrel pattern **806'** and leaves the spacer **808** standing over the dielectric layer **804**. Refer to FIG. **11A** (a top view of the device **162**), FIG. **11B** (a cross-sectional view of the device **162** along the 3-3 line of FIG. **11A**), and FIG. **11C** (cross-sectional views of the device **162** along the 2A-2A, 2B-2B, and 2C-2C lines of FIG. **11A**). The line 3-3 is the line 1-1 shifted from the mandrel pattern **806'** to the spacer **808**. The spacer **808** remains over the dielectric layer **804** after the mandrel pattern **806'** has been removed, e.g., by an etching process selectively tuned to remove the dielectric material **806** but not the spacer material. The etching process can be a wet etching, a dry etching, or a combination thereof.

(39) At operation **360**, the method **310** (FIG. 7) etches the substrate **802** to form a continuous fin line **810** in the substrate **802**. Refer to FIG. **12A** (a top view of the device **162**), FIG. **12B** (a cross-sectional view of the device **162** along the 3-3 line of FIG. **12A**), and FIG. **12C** (cross-sectional views of the device **162** along the 2A-2A, 2B-2B, and 2C-2C lines of FIG. **12A**). The fin line **810** includes multiple sections connected to form a continuous piece. For example, the fin line **810** includes a section **810A** in the region **104**, a section **810B** in the region **108**, a section **810C** in the region **106**, as well as other sections. To form the fin line **810**, the substrate **802** is etched with the spacer **808** as an etch mask. The spacer **808** and the dielectric layer **804** are subsequently removed. The etching process can be a wet etching, a dry etching, or a combination thereof.

(40) At operation **362**, the method **310** (FIG. 7) performs a fin cut process with the second mask manufactured in the operation **308** (FIG. 3). In the present embodiment, the second mask includes a pattern corresponding to the space **108** such as the pattern **116** of FIG. **6E**. The second mask may further include one or more patterns for cutting fin ends, such as the patterns **118** of FIG. **6E**, and one or more patterns for removing dummy fins. Refer to FIG. **13A** (a top view of the device **162**), FIG. **13B** (a cross-sectional view of the device **162** along the 3-3 line of FIG. **13A**), and FIG. **13C** (cross-sectional views of the device **162** along the 2A-2A, 2B-2B, and 2C-2C lines of FIG. **13A**). Two fins, **810A** and **810D**, are formed in the region **104**. Four fins, **810C**, **810F**, **810G**, and **810H**, are formed in the region **106**. The portions of the fin line **810** covered by the cut patterns (cut regions) are substantially removed. However, as shown in FIGS. **13B** and **13C**, small portions of the fin line **810** in the cut regions may remain because fin etching process typically does not completely etch to the bottom of the fin line **810** to avoid over-etching of the substrate **802**. The small residual portions of the fin line **810** are referred to as fin stubs in the following discussion because they are much shorter (along the Z direction) than the regular fins (e.g., **810A**). For example, the fin sections **810B** and **810E** have become fin stubs **810B** and **810E** after the fin cut process. With reference to FIG. **13A**, the fins **810A**, **810C**, **810D**, **810F**, **810G**, and **810H** are oriented lengthwise in the X direction; the fin stub **810B** is oriented lengthwise in a first direction different from the X direction; and the fin stub **810E** is oriented lengthwise in a second direction different from both the first direction and the X direction. The fin stub **810B** connects a bottom portion of the fins **810A** and **810C**. The fin stub **810E** connects a bottom portion of the fins **810D** and **810F**. There are other fin stubs shown in FIG. **13A**, though not labeled. In an embodiment, the operation **362** may also remove dummy fins, i.e. fins that are not used for forming transistors. For example, the second mask may include a cut pattern that removes the fin **810G**.

(41) In the present embodiment, the fin cut process includes a lithography process and an etching process. For example, a photoresist layer is formed on the silicon substrate using a spin-coating process and soft baking process. Then, the photoresist layer is exposed to a radiation using the second mask manufactured in the operation **308**. The exposed photoresist layer is subsequently developed and stripped thereby forming a patterned photoresist layer. The fin line **810** is partially protected by the patterned photoresist layer. Subsequently, the fin line **810** is etched through the openings of the patterned photoresist layer. The patterned photoresist layer is removed thereafter using a suitable process, such as wet stripping or plasma ashing.

(42) At operation **364**, the method **310** (FIG. 7) forms an isolation feature **812** over the substrate **802**. Refer to FIG. **14A** (a top view of the device **162**), FIG. **14B** (a cross-sectional view of the

device **162** along the 3-3 line of FIG. **14A**), and FIG. **14C** (cross-sectional views of the device **162** along the 2A-2A, 2B-2B, and 2C-2C lines of FIG. **14A**). The isolation feature **812** electrically isolate the various fins, including the fins **810A** and **810C**. Furthermore, the various fin stubs, including the fin stub **810B**, are buried underneath the isolation feature **812**. In an embodiment, operation **364** forms the isolation feature **812** by depositing a dielectric material such as silicon oxide over the substrate **802** and then etches back the dielectric material. In the present embodiment, portions of the fins **810** extending above the isolation feature **812** provide source, drain, and channel regions for FinFETs. For example, the fins **810A**, **810C**, **810D**, and **810H** partially extend above the isolation feature **812** and each provide source, drain, and channel region for one or more FinFETs.

(43) At operation **366**, the method **310** (FIG. **7**) performs further processes to complete the fabrication of the final IC device **162**. For example, the operation **366** may form source and drain regions in the fins (e.g., **810A** and **810C**) using ion implantation, epitaxial growth, and/or other suitable methods. For example, the operation **366** may form gate stacks over the fins (e.g., **810A** and **810C**) using a gate-first process or a gate-last process. Other processes include forming source and drain contacts, forming gate contacts, and forming via and metal interconnects, and so on.

(44) Although not intended to be limiting, the present disclosure provides many benefits to the manufacturing of an IC. For example, by connecting mandrel patterns in different layout blocks with dummy mandrel patterns, embodiments of the present disclosure reduce the space between the different layout blocks. This increases pattern density and reduces material costs per IC device. This also increases pattern density for improving fin uniformity, fin critical dimension, and CMP loading effect during various stages of fin etching processes. Further, embodiments of the present disclosure provide flexible schemes for inserting the dummy mandrel patterns, which may be tuned for specific process needs.

(45) In one exemplary aspect, the present disclosure is directed to a method. The method includes receiving an integrated circuit design layout that includes first and second layout blocks separated by a first space. The first and second layout blocks include, respectively, first and second line patterns oriented lengthwise in a first direction. The method further includes adding a dummy pattern to the first space, which connects the first and second line patterns. The method further includes outputting a mandrel pattern layout and a cut pattern layout in a computer-readable format. The mandrel pattern layout includes the first and second line patterns and the dummy pattern. The cut pattern layout includes a pattern corresponding to the first space. In embodiments, the method further includes manufacturing a first mask with the mandrel pattern layout and manufacturing a second mask with the cut pattern layout.

(46) In embodiments, the method further includes patterning a substrate with the first mask and the second mask. To further this embodiment, the method includes performing a first patterning process to a substrate with the first mask, thereby forming one or more features on the substrate, and performing a second patterning process to the substrate with the second mask, thereby removing a first portion of the one or more features, wherein the first portion lies in a region corresponding to the first space in the IC design layout.

(47) In another exemplary aspect, the present disclosure is directed to a method. The method includes receiving an integrated circuit (IC) design layout. The IC design layout includes a first layout block and a second layout block. The first layout block includes a first plurality of line patterns that are oriented lengthwise in a first direction and spaced from each other with a first pitch along a second direction that is orthogonal to the first direction. The second layout block includes a second plurality of line patterns that are oriented lengthwise in the first direction and spaced from each other with a second pitch along the second direction, and the first and second layout blocks are separated by a first space. The method further includes adding a dummy pattern to the first space, wherein the dummy pattern connects one of the first plurality and one of the second plurality. The method further includes outputting a mandrel pattern layout and a cut pattern

layout in a computer-readable format. The mandrel pattern layout includes the first and second pluralities and the dummy pattern. The cut pattern layout includes a pattern corresponding to the first space.

(48) In an embodiment, the method further includes adding another dummy pattern to the first space that connects the one of the first plurality and another one of the second plurality. In a further embodiment, the one of the second plurality is adjacent to the other one of the second plurality.

(49) In some embodiments, the first pitch is different from the second pitch. In some embodiments, each of the first plurality has a first line width, each of the second plurality has a second line width, and the first and second line widths are different.

(50) In another exemplary aspect, the present disclosure is directed to a semiconductor device. The semiconductor device includes a first fin on a substrate, wherein the first fin provides source, drain, and channel regions for a first field effect transistor (FET). The semiconductor device further includes a second fin on the substrate, wherein the second fin provides source, drain, and channel regions for a second FET. The semiconductor device further includes a first fin stub on the substrate, wherein the first fin stub connects a bottom portion of the first fin and a bottom portion of the second fin. The semiconductor device further includes an isolation feature over the first fin stub and between the first and second fins. From a top view, the first and second fins are oriented lengthwise in a first direction, and the first fin stub is oriented lengthwise in a second direction that is different from the first direction. In embodiments, the first fin lies completely in a first rectangular region, the second fin lies completely in a second rectangular region, and the first and second rectangular regions are arranged side by side along the first direction.

(51) In an embodiment, the semiconductor device further includes a third fin, a fourth fin, and a second fin stub. The third fin provides source, drain, and channel regions for a third FET. The fourth fin provides source, drain, and channel regions for a fourth FET. The second fin stub connects a bottom portion of the third fin and a bottom portion of the fourth fin. From a top view, the third and fourth fins are oriented lengthwise in the first direction, and the second fin stub is oriented lengthwise in a third direction that is different from the first and second directions. In a further embodiment, the third fin lies completely in the first rectangular region, and the fourth fin lies completely in the second rectangular region.

(52) The foregoing outlines features of several embodiments so that those having ordinary skill in the art may better understand the aspects of the present disclosure. Those having ordinary skill in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those having ordinary skill in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a substrate; a semiconductor fin on the substrate, the semiconductor fin oriented lengthwise in a first direction; a residual feature on the substrate and abutting a first end of the semiconductor fin, the residual feature lower than the semiconductor fin in height, the residual feature and the semiconductor fin including a same material composition, the residual feature oriented lengthwise in a second direction that is different from the first direction; and an isolation feature covering the residual feature and over sidewalls of the semiconductor fin.
2. The semiconductor device of claim 1, wherein the residual feature and the semiconductor fin include silicon.
3. The semiconductor device of claim 1, wherein the first and second directions form an angle not more than 45 degrees.

4. The semiconductor device of claim 1, wherein the residual feature is a first residual feature, the semiconductor device further comprising: a second residual feature on the substrate and abutting a second end of the semiconductor fin, the second end opposing the first end of the semiconductor fin, the second residual feature lower than the semiconductor fin in height, the first and second residual features including the same material composition.
5. The semiconductor device of claim 4, wherein the second residual feature is oriented lengthwise in a third direction that is different from the first direction.
6. The semiconductor device of claim 4, wherein the second residual feature is oriented lengthwise in a third direction that is different from the second direction.
7. The semiconductor device of claim 1, wherein the semiconductor fin is a first semiconductor fin and the residual feature is a first residual feature, the semiconductor device further comprising: a second semiconductor fin on the substrate, the second semiconductor fin oriented lengthwise in the first direction; a third semiconductor fin on the substrate, the third semiconductor fin oriented lengthwise in the first direction; and a second residual feature connecting a bottom portion of the second semiconductor fin to a bottom portion of the third semiconductor fin, the second residual feature lower than the second and third semiconductor fins in height.
8. The semiconductor device of claim 7, wherein the second residual feature oriented lengthwise in a third direction that is different from the first direction.
9. The semiconductor device of claim 7, wherein the second residual feature oriented lengthwise in a tilted angle with respect to the second direction.
10. The semiconductor device of claim 7, further comprising: a fourth semiconductor fin on the substrate, the fourth semiconductor fin oriented lengthwise in the first direction; a fifth semiconductor fin on the substrate; the fifth semiconductor fin oriented lengthwise in the first direction; and a third residual feature connecting a bottom portion of the fourth semiconductor fin to a bottom portion of the fifth semiconductor fin, the third residual feature lower than the fourth and fifth semiconductor fins in height.
11. The semiconductor device of claim 1, wherein the first and second directions form an angle not more than 30 degrees.
12. A semiconductor device, comprising: a substrate; first, second, third semiconductor fins on the substrate, lengthwise directions of the first, second, and third semiconductor fins parallel to each other; a first residual feature on the substrate and abutting a first end of the first semiconductor fin, the first residual feature lower than the first semiconductor fin in height, a lengthwise direction of the first residual feature tilted with respect to a lengthwise direction of the first semiconductor fin; and a second residual feature on the substrate and connecting a bottom portion of the second semiconductor fin and to a bottom portion of the third semiconductor fin, the second residual feature lower than the second and third semiconductor fins in height, a lengthwise direction of the second residual feature tilted with respect to the lengthwise direction of the first residual feature, wherein the first, second, third semiconductor fins and the first and second residual features include a same material composition.
13. The semiconductor device of claim 12, wherein the same material composition is silicon.
14. The semiconductor device of claim 12, wherein the lengthwise direction of the second residual feature is tilted with respect to the lengthwise direction of the first semiconductor fin.
15. The semiconductor device of claim 12, further comprising: an isolation feature covering the first and second residual features and over sidewalls of the first, second, and third semiconductor fins.
16. The semiconductor device of claim 12, further comprising: a third residual feature on the substrate and abutting a second end of the first semiconductor fin, the third residual feature lower than the first semiconductor fin in height, a lengthwise direction of the third residual feature tilted with respect to the lengthwise direction of the first residual feature.
17. A semiconductor device, comprising: a substrate; a semiconductor fin protruding from the

substrate; a first residual feature protruding from the substrate and abutting a first end of the semiconductor fin, the first residual feature lower than the semiconductor fin in height, a lengthwise direction of the first residual feature tilted from a lengthwise direction of the semiconductor fin; a second residual feature protruding from the substrate and abutting a second end of the semiconductor fin, the second residual feature lower than the semiconductor fin in height, a lengthwise direction of the second residual feature tilted from the lengthwise direction of the first residual feature; and an isolation feature covering the first and second residual features and over sidewalls of the semiconductor fin, wherein the first and second residual features and the semiconductor fin include a same material composition.

18. The semiconductor device of claim 17, wherein the lengthwise direction of the first residual feature is tilted from the lengthwise direction of the semiconductor fin for an angle no more than 45 degrees.

19. The semiconductor device of claim 17, wherein the lengthwise direction of the first residual feature is tilted from the lengthwise direction of the semiconductor fin for an angle no more than 30 degrees.

20. The semiconductor device of claim 17, wherein the lengthwise direction of the second residual feature is perpendicular to the lengthwise direction of the semiconductor fin.
